

ASAP: Purification and crystallization of EV_A71_2A precursor

PAGE24-00801

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Constructs info

A71EV2A-c058

MCSSHHHHHHGSGSGSDQEAKPSTEDLGDKKEGEYIKLVIGQDSSEIHFVKVMTTHLKKLKESYQQRQGVPMNSLRFLFEG

A71EV2A-p018

Protein precipitated during overnight dialysis, Ni-rebinding and concentration

Protein stick to Ni beads during Ni-rebinding

Sumo tag contamination problem ('higher band' belong to target protein)

Purification

Resuspend pellet in Ni²⁺ binding buffer: ~30ml of pellet + 120ml of buffer for 1L

- sonication at 5sec on/ 10 sec off at 35% for total 25 mins

- transfer for high speed centrifuge: 50ml tube, 20,000g for 1h

Ni²⁺ IMAC

binding buffer: 50 mM HEPES, pH 7.5

500 mM NaCl

20 mM Imidazole, pH 7.0

5% glycerol

1 mM TCEP

elution buffer: 50 mM HEPES, pH 7.5

500 mM NaCl

300 mM Imidazole, pH 7.0

5% glycerol

1 mM TCEP

- ~ 4-5 ml of Ni Beads

- Loading supernatant and wash with full column volume (120ml) of binding buffer supplemented with 20mM Imidazole

- wash beads with 30ml of 30mM imidazole

- Elute with 30 ml of elution buffer (containing 300mM Imidazole)

-measure A280. Add protease (sumo) 1OD unit Sumo for every 50OD units target, incubate on 4°C overnight

Dialysis buffer:

50mM HEPES, pH7.0

350mM NaCl

0.5mM TCEP

5% Glycerol

Ni-rebinding: pre-equilibrate with base buffer containing 20mM imidazole

Ni beads: double columns- 2ml + 1ml

wash Ni-beads with base buffer containing 20mM, 50mM and 300mM imidazole 20 ml each (target protein stick to Ni beads, see SDS-PAGE bellow)

C058 Protein slightly precipitated after overnight dialysis (initial concertation at 2.5mg/ml)

Gel Filtration

- Buffer:

25mM HEPES pH7.5

200mM NaCl

0.5mM TCEP

5% Glycerol

Concentration & mass spec check

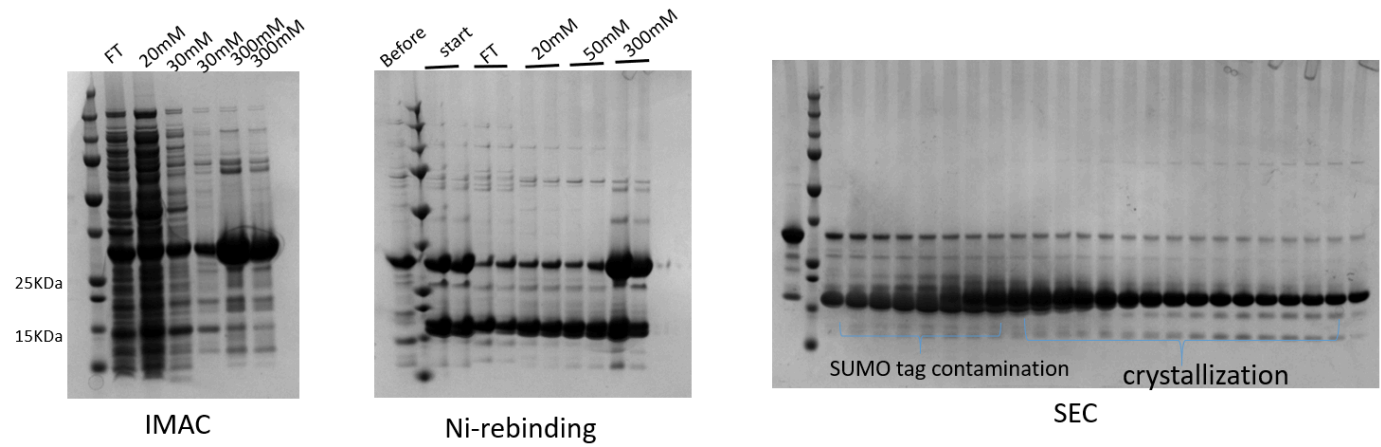
- Pooling GF peak, fractions:

- Using 10 kDa cut off

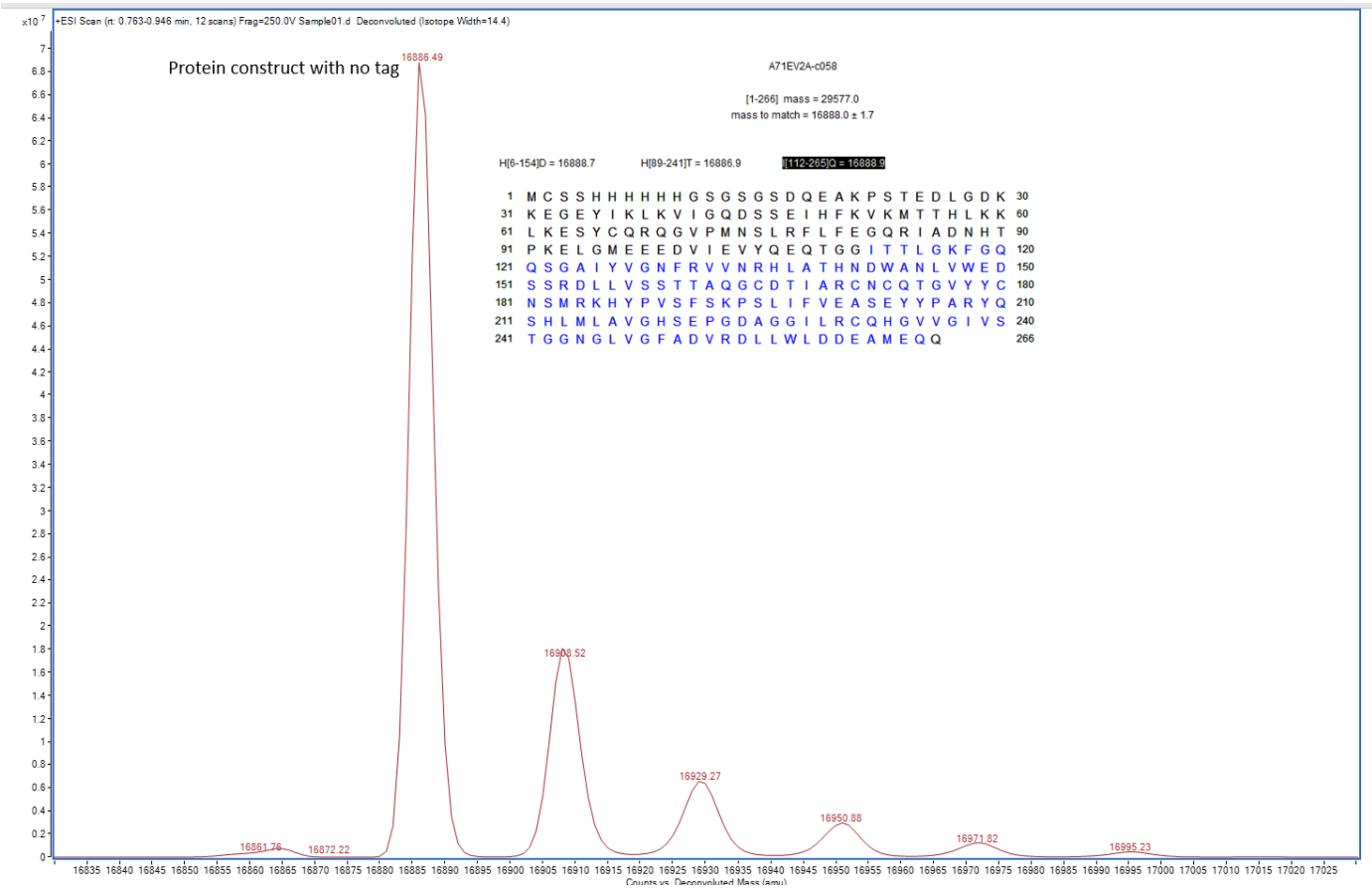
- Concentration check

Protein concentrated to 17mg/ml for crystallization

SDS-PAGEs



MS



Crystallization

20240501	Plate Barcode	Construct ID	concentration (mg/ml)	Screen	Temperature
25mM HEPES pH7.5	CI087611	A71EV2A-c058	17	LFS	20
200mM NaCl	CI087612	A71EV2A-c058	17	HCS	20
0.5mM TCEP	CI087613	A71EV2A-c058	17	HIN	20
5% Glycerol	CI087614	A71EV2A-c058	17	JCSG	20
	CI087615	A71EV2A-c058	17	SG1	20
	CI087616	A71EV2A-c058	17	SG1-2	20